Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If A and B are two events. The probability that at most one of A, B occurs, is

(A) $1 - P(A \cap B)$

(B) $P(\bar{A}) + P(\bar{B}) - P(\bar{A} \cap \bar{B})$

(C) $P(\bar{A}) + P(\bar{B}) + P(A \cup B) - 1$

(D) All of these

Q.2

In a series of 3 one-day cricket matches between teams A and B of a college, the probability of team A winning or drawing are $1/3$ and $1/6$ respectively. If a win, loss or draw gives 2,0 and 1 point respectively, then what is the probability that team A will score 5 points in the series?

(A) $\frac{17}{18}$

(B) $\frac{11}{12}$

(C) $\frac{1}{12}$

(D) $\frac{1}{18}$

Q.3

If $\mathrm{A}, \mathrm{B}, \mathrm{C}$ are events such that $\mathrm{P}(\mathrm{A}) = 0.3, \mathrm{P}(\mathrm{B}) = 0.4, \mathrm{P}(\mathrm{C}) = 0.8, \mathrm{P}(\mathrm{A} \cap \mathrm{B}) = 0.08, \mathrm{P}(\mathrm{A} \cap \mathrm{C}) = 0.28$ $\mathrm{P}(\mathrm{A} \cap \mathrm{B} \cap \mathrm{C}) = 0.09$ If $\mathrm{P}(\mathrm{A} \cup \mathrm{B} \cup \mathrm{C}) \geq 0.75$, then find the range of $x = P(B \cap C)$ lies in the interval

(A) $0.23 \leq x \leq 0.48$

(B) $0.23 \leq x \leq 0.47$

(C) $0.22 \leq \mathrm{x} \leq 0.48$

(D) None of these

Q.4

A machine has three parts, A, B and C, whose chances of being defective are $0.02, 0.10$ and 0.05 respectively. The machine stops working if any one of the parts becomes defective. What is the probability that the machine will not stop working?

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(A) 0.06

(B) 0.16

(C) 0.84

(D) 0.94

Q.5

For two mutually exclusive events A and B, $\mathrm{P}(\mathrm{A}) = 0.2$ and $\mathrm{P}(\overline{\mathrm{A}} \cap \mathrm{B}) = 0.3$. What is $\mathrm{P}(\mathrm{A} \mid (\mathrm{A} \cup \mathrm{B}))$ equal to?

(A) $\frac{1}{2}$

(B) $\frac{2}{5}$

(C) $\frac{2}{7}$

(D) $\frac{2}{3}$

Q.6

An aircraft has three engines A, B and C. The aircraft crashes if all the three engines fail. The probabilities of failure are $0.03, 0.02$ and 0.05 for engines A, B and C respectively. What is the probability that the aircraft will not crash?

(A) 0.00003

(B) 0.90

(C) 0.99997

(D) 0.90307

Q.7

Let A, B, C be three events. If the probability of occurring exactly one event out of A and B is $1 - a$, out of B and C and A is $1 - a$ and that of occurring three events simultaneously is a^2, then the probability that at least one out of A, B, C will occur is

(A) $1/2$

(B) Greater than $1/2$

(C) Less than $1/2$

(D) Greater than $3/4$

Q.8

From past experience it is known that an investor will invest in security A with a probability of 0.6, will invest in security B with a probability 0.3 and will invest in both A and B with a probability of 0.2. What is the probability that an investor will invest neither in A nor in B?

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(A) 0.7

(B) 0.28

(C) 0.3

(D) 0.4

Q.9

For the three events A, B and C, P (exactly one of the events A or B occurs $)=P$ (exactly one of the two events B or C occurs $)=P($ exactly one of the events C or A occurs $)=p$ and $P($ all the three events occur simultaneously$)=p^2$, where $0<p<1/2$. Then the probability of at least one of the three events A, B and C occurring is

(A) $\frac{3p+2p^2}{2}$

(B) $\frac{p+3p^2}{4}$

(C) $\frac{p+3p^2}{2}$

(D) $\frac{3p+2p^2}{4}$

Q.10

Let A and B be two events. Then $1+P(A \cap B)-P(B)-P(A)$ is equal to

(A) $P(\bar{A} \cup \bar{B})$

(B) $P(A \cap \bar{B})$

(C) $P(\bar{A} \cap B)$

(D) $P(\bar{A} \cap \bar{B})$

Q.11

Let E^c denote the complement of an event E. Let E, F, G be pairwise independent events with $\mathrm{P}(\mathrm{G})>0$ and $\mathrm{P}(\mathrm{E} \cap \mathrm{F} \cap \mathrm{G})=0$. Then $\mathrm{P}\left(\mathrm{E}^{\mathrm{c}} \cap \mathrm{F}^{\mathrm{c}} \mid \mathrm{G}\right)$ equals

(A) $\mathrm{P}\left(\mathrm{E}^{\mathrm{c}}\right)+\mathrm{P}\left(\mathrm{F}^{\mathrm{c}}\right)$

(B) $\mathrm{P}\left(\mathrm{E}^{\mathrm{c}}\right)-\mathrm{P}\left(\mathrm{F}^{\mathrm{c}}\right)$

(C) $\mathrm{P}\left(\mathrm{E}^{\mathrm{c}}\right)-\mathrm{P}(\mathrm{F})$

(D) $\mathrm{P}(\mathrm{E})-\mathrm{P}\left(\mathrm{F}^{\mathrm{c}}\right)$

Q.12

If X follows a binomial distribution with parameters $n=8$ and $\mathrm{p}=\frac{1}{2}$, then $\mathrm{P}(|\mathrm{X}-4| \leq 2)$ is

(A) $\frac{119}{128}$

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(B) $\frac{119}{228}$

(C) $\frac{19}{128}$

(D) $\frac{18}{128}$

Q.13

A bag contains n balls. It is given that the probability that among these n balls exactly r balls are white is proportional to $r^2(0 \leq r \leq n)$. A ball is drawn at random and is found to be white. Then the probability that all the balls in the bag are white, will be:

(A) $\frac{2n}{(n+1)^2}$

(B) $\frac{4n}{(n+1)^2}$

(C) $\frac{2n}{(n+3)^2}$

(D) $\frac{4n}{(n+3)^2}$

Q.14

A student appears for tests I, II and III. The student is successful if he passes either in tests I and II or tests I and III. The probabilities of the student passing in tests I, II, III are p, q and $\frac{1}{2}$ respectively. The probability that the student is successful is $\frac{1}{2}$ then the relation between p and q is given by

(A) $\mathrm{pq}+\mathrm{p}=1$

(B) $\mathrm{p}^2+\mathrm{q}=1$

(C) $\mathrm{pq}-1=\mathrm{p}$

(D) None of these

Q.15

A man takes a step forward with probability 0.4 and backward with probability 0.6. The probability that at the end of eleven steps he is one step away from the starting point is

(A) $\frac{2^5 \cdot 3^5}{5^{10}}$

(B) $462 \times \left(\frac{6}{25}\right)^5$

(C) $231 \times \frac{3^5}{5^{10}}$

(D) None of these

Q.16

If E_1 and E_2 are two events such that $\mathrm{P}\left(\mathrm{E}_1\right)=1/4\ \mathrm{P}\left(\mathrm{E}_2/\mathrm{E}_1\right)=1/2$ and $\mathrm{P}\left(\mathrm{E}_1/\mathrm{E}_2\right)=1/4$, then choose the incorrect
statement.

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(A) E_1 and E_2 are independent

(B) E_1 and E_2 are exhaustive

(C) E_2 is twice as likely to occur as E_1

(D) Probabilities of the events $E_1 \cap E_2$, E_1 and E_2 are in GP.

Q.17

If $\frac{1+4p}{4}$, $\frac{1-p}{2}$ and $\frac{1-2p}{2}$ are the probabilities of three mutually exclusive events, then value of p is

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{1}{4}$

(D) $\frac{2}{3}$

Q.18

If A and B are two events, odds against A are $2:1$ and odds in favour of $A \cup B$ are $3:1$, and the range of values of P (B) is $[x, \quad y]$, then the value of $\frac{1}{y-x}$ is

(A) 1

(B) 2

(C) 3

(D) 4

Q.19

Let A and B be two events such that $P(\overline{A \cup B}) = \frac{1}{6}, P(A \cap B) = \frac{1}{4}$ and $P(\bar{A}) = \frac{1}{4}$, where $\bar{A}$ stands for component of event A. Then events A and B are

(A) independent but not equally likely

(B) mutually exclusive and independent

(C) equally likely and mutually exclusive

(D) equally likely but not independent

Q.20

A die is thrown. Let A be the event that the number obtained is greater than 3 . Let B be the event that the number obtained is less than 5 . Then $P(A \cup B)$ is

(A) 0

(B) 1

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(C) $2/5$

(D) $3/5$

Answers & Solutions

Q.1 Answer:

All of these

Solution:

Required probability $= P(\bar{A} \cup \bar{B}) = P(A \cap B)$
$= 1 - P(\mathrm{A} \cap \mathrm{B})$
Again, $P(\bar{A} \cup \bar{B}) = P(\bar{A}) + P(\bar{B}) - P(\bar{A} \cap \bar{B})$
[By add. Theorem] Again, $P(\bar{A} \cup \bar{B}) = P(\bar{A}) + P(\bar{B}) - P(\bar{A} \cap \bar{B})$
$= P(\bar{A}) + P(\bar{B}) - P\overline{(A \cup B)}$
$= P(\bar{A}) + P(\bar{B}) - \{1 - P(A \cup B)\}$
$= P(\bar{A}) + P(\bar{B}) + P(A \cup B) - 1$
Finally, $P(\bar{A} \cup \bar{B}) = P[(A \cap \bar{B}) \cup (\bar{A} \cap B) \cup (\bar{A} \cap \bar{B})]$
$= P(A \cap \bar{B}) + P(\bar{A} \cap B) + P(\bar{A} \cap \bar{B})$
$[\because A \cap \bar{B}, \bar{A} \cap B$ **and** $\bar{A} \cap \bar{B}$ **are mutually**
exclusive events] So, alternative (d) **is the correct answer.**

Q.2 Answer:

$\frac{1}{18}$

Solution:

Req. Prob. $= \mathrm{P}(5$ **points** $) = \mathrm{P}($ **two wins and one draw)** $= \mathrm{P}(\mathrm{WWD}) + \mathrm{P}(\mathrm{WDW}) + \mathrm{P}(\mathrm{DWW})$
$\frac{1}{3} \times \frac{1}{3} \times \frac{1}{6} + \frac{1}{3} \times \frac{1}{6} \times \frac{1}{3} + \frac{1}{6} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{18}$

Q.3 Answer:

$0.23 \leq x \leq 0.48$

Solution:

Since $\mathrm{P}(\mathrm{A} \cup \mathrm{B} \cup \mathrm{C}) = \mathrm{P}(\mathrm{A}) + \mathrm{P}(\mathrm{B}) + \mathrm{P}(\mathrm{C})$
$-\mathrm{P}(\mathrm{A} \cap \mathrm{B}) - \mathrm{P}(\mathrm{B} \cap \mathrm{C}) - \mathrm{P}(\mathrm{C} \cap \mathrm{A}) + \mathrm{P}(\mathrm{A} \cap \mathrm{B} \cap \mathrm{C})$
Or $\mathrm{P}(\mathrm{A} \cup \mathrm{B} \cup \mathrm{C}) = 0.3 + 0.4 + 0.8 - (0.08 + 0.28$
$+\mathrm{P}(\mathrm{B} \cap \mathrm{C})) + 0.09$
$\mathrm{F} = 1.23 - \mathrm{P}(\mathrm{B} \cap \mathrm{C})$
Or $\mathrm{P}(\mathrm{B} \cap \mathrm{C}) = 1.23 - \mathrm{P}(\mathrm{A} \cup \mathrm{B} \cup \mathrm{C})$
But we know that $0 \leq \mathrm{P}(\mathrm{A} \cup \mathrm{B} \cup \mathrm{C}) \leq 1$
Hence $0.23 \leq \mathrm{P}(\mathrm{B} \cap \mathrm{C}) \leq 0.48$

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Q.4 Answer:

0.84

Solution:

Probability that machine stops working $= P(A \cup B \cup C)$

$\Rightarrow P(A \cup B \cup C) = P(A) + P(B) + P(C)$

$-P(A \cap B) - P(A \cap C) - P(B \cap C)$

$+P(A \cap B \cap C)$

$\Rightarrow P(A \cup B \cup C) = P(A) + P(B) + P(C)$

$-P(A)P(B) - P(A)P(C)$

$-P(B)P(C) + P(A)P(B)P(C)$

$(\because A, B \& C$ **are independent events** $) \Rightarrow P(A \cup B \cup C) = 0.02 + 0.1 + 0.05 - (0.02 \times 0.1)$

$-(0.02 \times 0.05) - (0.1 \times 0.05)$

$+(0.02 \times 0.05 \times 0.1)$

$\Rightarrow P(A \cup B \cup C) = 0.16$

$\therefore$ **Probability that the machine will not stop working** $= 1 - P(A \cup B \cup C) = 1 - 0.16 = 0.84$

Q.5 Answer:

$\frac{2}{5}$

Solution:

Events A **and** B **are mutually excusive,**

Hence $P(A \cap B) = \phi = 0$

$\therefore P(A \cap B) = P(A) + P(B) \quad \ldots.(i)$

$P(A) = 0.2$ [given]

$P(B) = P(\bar{A} \cap B) + P(A \cap B)$

$P(B) = P(\bar{A} \cap B) \quad [\because P(A \cap B) = 0]$

$= 0.3$

$P(A \cap B) = 0.2 + 0.3 = 0.5$

$P(A \mid A \cap B) = \frac{P(A)}{P(A \cup B)} = \frac{0.2}{0.5} = \frac{2}{5}$

$P(A \mid (A \cup B)) = \frac{2}{5}$

Q.6 Answer:

0.99997

Solution:

Since, probabilities of failure for engines A, B and $CP(A), P(B)$ **and** $P(C)$ **are** $0.03, 0.02$ **and** 0.05 **respectively. The aircraft will crash only when all the three engine fail. So, probability that it crashes** $= \mathrm{P(A)} \cdot \mathrm{P(B)} \cdot \mathrm{P(C)}$

$= 0.03 \times 0.02 \times 0.05$

$= 0.00003$

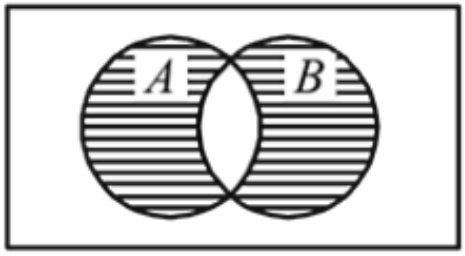
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Hence, the probability that the aircraft will not crash $= 1 - 0.00003$
$= 0.99997$

Q.7 Answer:

Greater than $1/2$

Solution:



$P($ **exactly one event out of** A **and** B **occurs)** $= P\left[A \cap B'\right) \cup (A' \cap B)]$
$= P(A \cup B) - P(A \cap B)$
$= P(A) + P(B) - 2P(A \cap B)$
$\therefore P(A) + P(B) - 2P(A \cap B) = 1 - a$ **...(1)**
Similarly, $P(B) + P(C) - 2P(B \cap C) = 1 - 2a$ **...(2)**
$P(C) + P(A) - 2P(C \cap A) = 1 - a$
$P(A \cap B \cap C) = a^2$ **...(3)**
Now $P(A \cup B \cup C)$
$= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C)$
$-P(C \cap A) + P(A \cap B \cap C)$
$= \frac{1}{2}[P(A) + P(B) - 2P(B \cap C) + P(B) + P(C)$
$-2P(B \cap C) + P(C) + P(A) - 2P(C \cap A)]$
$+P(A \cap B \cap C)$
$= \frac{1}{2}[1 - a + 1 - 2a + 1 - a] + a^2$
[**using** $(1), (2), (3)$] **and** $= \frac{3}{2} - 2a + a^2 = \frac{1}{2} + (a-1)^2 > \frac{1}{2}$

Q.8 Answer:

0.3

Solution:

As given $\mathrm{P(A)} = 0.6, \mathrm{P(B)} = 0.3$ **and**
$\mathrm{P(A \cap B)} = 0.2$
$\mathrm{P(A \cup B)} = \mathrm{P(A)} + \mathrm{P(B)} - \mathrm{P(A \cap B)}$
$= 0.6 + 0.3 - 0.2 = 0.9 - 0.2 = 0.7$
So, $\mathrm{P}($ **neither in** A **nor in** $\mathrm{B}) = 1 - \mathrm{P(A \cup B)}$
$= 1 - 0.7 = 0.3$

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Q.9 Answer:

$\frac{3p+2p^2}{2}$

Solution:

We know that $P($ **exactly one of** A **or** B **occurs)** $= P(A) + P(B) - 2P(A \cap B)$

$\therefore P(A) + P(B) - 2P(A \cap B) = p$ **...(1)**

Similarly, $P(B) + P(C) - 2P(B \cap A) = p \ldots (2)$

and $P(C) + P(A) - 2P(C \cap A) = p$

Adding Eqs. (1), (2) and (3) we get $2[P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C)$

$-P(C \cap A)] = 3p$

$\Rightarrow P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C)$

$-P(C \cap A)] = 3p/2$ **...(3)**

It is also given that $P(A \cap B \cap C) = p^2$ **...(4)**

Now, $P($ **at least one of** A, B **and** C **)** $= \mathrm{p(A) + p(B) + p(C) - p(A \cap B) - p(B \cap C)}$

$-\mathrm{p(C \cap B) + p(A \cap B \cap C)}$

$= \frac{3p}{2} + p^2 = \frac{3p+2p^2}{2}$ **[Using Eqs. (4) and (5).]**

Q.10 Answer:

$P(\bar{A} \cap \bar{B})$

Solution:

We know $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Consider; $1 + P(A \cap B) - P(B) - P(A)$

$= 1 - P(B) - P(A) + P(A) + P(B) - P(A \cup B)$

$= 1 - P(A \cup B)$

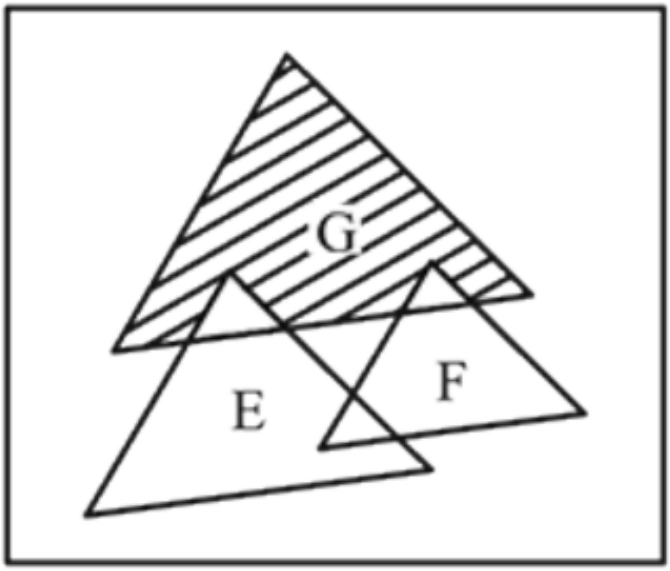
Q.11 Answer:

$\mathrm{P\,(E^c) - P(F)}$

Solution:

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We have



$\because \mathrm{E} \cap \mathrm{F} \cap \mathrm{G}=\phi$

$\mathrm{P}\left(\mathrm{E}^{c} \cap \mathrm{F}^{c} / \mathrm{G}\right)=\frac{\mathrm{P}\left(\mathrm{E}^{c} \cap \mathrm{F}^{c} \cap \mathrm{G}\right)}{\mathrm{P}(\mathrm{G})}$

$=\frac{P(G)-P(E \cap G)-P(G \cap F)}{P(G)}$

[**From venn diagram** $\mathrm{E}^{\mathrm{c}} \cap \mathrm{F}^{\mathrm{c}} \cap \mathrm{G}=\mathrm{G}-\mathrm{E} \cap \mathrm{G}-\mathrm{F} \cap \mathrm{G}$]

$=\frac{\mathrm{P}(\mathrm{G})-\mathrm{P}(\mathrm{E}) \mathrm{P}(\mathrm{G})-\mathrm{P}(\mathrm{G}) \mathrm{P}(\mathrm{F})}{\mathrm{P}(\mathrm{G})}$

[$\because$ **E, F, G are pairwise independent**] $=1-\mathrm{P}(\mathrm{E})-\mathrm{P}(\mathrm{F})=\mathrm{P}\left(\mathrm{E}^{\mathrm{c}}\right)-\mathrm{P}(\mathrm{F})$

Q.12 Answer:

$\frac{119}{128}$

Solution:

We have, $\mathrm{P}(|\mathrm{X}-4| \leq 2)=\mathrm{P}(-2 \leq \mathrm{X}-4 \leq 2)=\mathrm{P}(2 \leq \mathrm{X} \leq 6)$

$=1-[\mathrm{P}(\mathrm{X}=0)+\mathrm{P}(\mathrm{X}=1)+\mathrm{P}(\mathrm{X}=7)+\mathrm{P}(\mathrm{X}=8)]$

$=1-\left[{}^{8} \mathrm{C}_{0}\left(\frac{1}{2}\right)^{8}+{}^{8} \mathrm{C}_{1}\left(\frac{1}{2}\right)^{8}+{}^{8} \mathrm{C}_{7}\left(\frac{1}{2}\right)^{8}+{}^{8} \mathrm{C}_{8}\left(\frac{1}{2}\right)^{8}\right.$

$=1-\left(\frac{1}{2}\right)^{8}(1+8+8+1)=1-\frac{18}{2^{8}}=\frac{119}{128}$

Q.13 Answer:

$\frac{4n}{(n+1)^2}$

Solution:

Let $\mathrm{E}_{i}=0,1,2, \ldots n$ **be the event that the bag contains exactly** i **white balls then** $\mathrm{p}\left(\mathrm{E}_{i}\right) k^{2} i$

where $\sum_{i=0}^{n} \mathrm{k}_{i}^{2}=1 \Rightarrow k=\frac{r}{n(n+1)(2n+1)}$

Let A **be the event that a ball drawn is white**

$p\left(\frac{E_{n}}{A}\right)=\frac{p(E_n) p(A / E_n)}{\sum_{i=0}^{n} k_{i}^{2}\left(\frac{i}{n}\right)}=\frac{K \cdot n^{2} \cdot 1}{\frac{K}{n} \sum_{t=0}^{n}(i)^{3}}=\frac{n^{3}}{\left(\frac{n(n+1)}{2}\right)^{2}}$

$p\left(\frac{\mathrm{E}_{n}}{\mathrm{A}}\right)=\frac{4n}{(n+1)^2}$

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Q.14 Answer:

$pq + p = 1$

Solution:

Let A, B and C be the events that the student is successful in tests I, II and III respectively. Then P (The student is successful) $= P(A)P(B)\{1 - P(C)\} + P(A)\{1 - P(B)\}P(C)$
$+P(A)P(B)P(C)$
$= p \cdot q\left(1 - \frac{1}{2}\right) + p(1 - q)\frac{1}{2} + p \cdot q\frac{1}{2}$
$= \frac{1}{2}pq + \frac{1}{2}p(1 - q) + \frac{1}{2}pq$
$= \frac{1}{2}(pq + p - pq + pq) = \frac{1}{2}(pq + p)$
$\therefore \frac{1}{2} = \frac{1}{2}(pq + p) \Rightarrow 1 = pq + p$

Q.15 Answer:

$462 \times \left(\frac{6}{25}\right)^5$

Solution:

As $0.4 + 0.6 = 1$, the man either takes a step forward or a step backward. Let a step forward be a success and a step backward be a failure. Then, the probability of success in one step $= p = 0.4 = \frac{2}{5}$
The probability of failure in one step $= q = 0.6 = \frac{3}{5}$**. In 11 steps he will be one step away from the starting point if the numbers of successes and failures differ by 1 So, the number of successes** $= 6$
The number of failures $= 5$ **or the number of successes** $= 5$**, The number of failures** $= 6 \therefore$ **the required probability** $= {}^{11}C_6 p^6 q^5 + {}^{11}C_5 p^5 q^6$
$= {}^{11}C_6 \left(\frac{2}{5}\right)^6 \cdot \left(\frac{3}{5}\right)^5 + {}^{11}C_5 \left(\frac{2}{5}\right)^5 \cdot \left(\frac{3}{5}\right)^6$
$= \frac{11!}{6!5!} \cdot \left(\frac{2}{5}\right)^5 \cdot \left(\frac{3}{5}\right)^5 \left\{\frac{2}{5} + \frac{3}{5}\right\}$
$= \frac{11.10.9.8.7}{120} \cdot \frac{2^5 \cdot 3^5}{5^{10}} = 462 \times \left(\frac{6}{25}\right)^5$

Q.16 Answer:

E_1 and E_2 are independent

Solution:

$P(E_2/E_1) = \frac{P(E_1 \cap E_2)}{P(E_1)} \Rightarrow \frac{1}{2} = \frac{P(E_1 \cap E_2)}{1/4}$
$\Rightarrow P(E_1 \cap E_2) = \frac{1}{8} = P(E_2) \cdot P(E_1/E_2) = P(E_2) \cdot \frac{1}{4}$
$\Rightarrow P(E_2) = \frac{1}{2}$
Since $P(E_1 \cap E_2) = \frac{1}{8} = P(E_1) \cdot P(E_2)$
$\Rightarrow$ **events are independent Also** $P(E_1 \cup E_2) = \frac{1}{2} + \frac{1}{4} - \frac{1}{8} = \frac{5}{8}$
$\Rightarrow E_1 \& E_2$ **are non-exhaustive**

Q.17 Answer:

$\frac{1}{2}$

Solution:

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$\frac{1+4p}{4}, \frac{1-p}{2}, \frac{1-2p}{2}$ **are probabilities of the three mutually exclusive events, then** $0 \leq \frac{1+4p}{4} \leq 1, 0 \leq \frac{1-p}{2} \leq 1, 0 \leq \frac{1-2p}{2} \leq 1$ **and** $0 \leq \frac{1+4p}{4} + \frac{1-p}{2} + \frac{1-2p}{2} \leq 1$

$\therefore -\frac{1}{4} \leq p \leq \frac{3}{4}, -1 \leq p \leq 1, -\frac{1}{2} \leq p\frac{1}{2}, \frac{1}{2} \leq p \leq \frac{5}{2}$

$\therefore \frac{1}{2} \leq \mathrm{p} \leq \frac{1}{2}$

[The intersection of above four intervals]

$\therefore \mathrm{p} = \frac{1}{2}$

Q.18 Answer:

3

Solution:

$P\left(\bar{A}\right) = \frac{2}{3} \Rightarrow P\left(A\right) = \frac{1}{3}$

$P\left(A \cup B\right) = \frac{3}{4}$

$\mathrm{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$

$\Rightarrow \frac{3}{4} = \frac{1}{3} + \mathrm{P}\left(\mathrm{B}\right) - \mathrm{P}\left(\mathrm{A} \cap \mathrm{B}\right)$

$\Rightarrow P\left(B\right) - P\left(A \cap B\right) = \frac{3}{4} - \frac{1}{3} = \frac{5}{12}$

$\Rightarrow P\left(B\right) \geq \frac{5}{12}$ ... i

Also $P(A) \geq P(A \cap B)$

$\Rightarrow P\left(A\right) \geq P\left(B\right) - \frac{5}{12}$

$\Rightarrow P\left(B\right) \leq P\left(A\right) + \frac{5}{12}$

$\Rightarrow P\left(B\right) \leq \frac{1}{3} + \frac{5}{12}$

$\Leftrightarrow P\left(B\right) \leq \frac{9}{12} = \frac{3}{4}$ ... ii

i **and** ii $\Rightarrow \frac{5}{12} \leq P\left(B\right) \leq \frac{3}{4}$

$\Rightarrow x = \frac{5}{12}, \quad y = \frac{3}{4} \quad \Rightarrow y - x = \frac{3}{4} - \frac{5}{12}$

$= \frac{4}{12} = \frac{1}{3}$

$\Rightarrow \frac{1}{y - x} = 3$

Q.19 Answer:

independent but not equally likely

Solution:

$\mathrm{P(A)} = \frac{3}{4}, \mathrm{P(A \cup B)} = \frac{5}{6}, \mathrm{P(A \cap B)} = \frac{1}{4}$

$\Rightarrow \mathrm{P(B)} = \frac{1}{3}$

So, $P(A \cap B) = P(A)P(B)$

Q.20 Answer:

1

Solution:

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$A = \{x : x > 3\} = \{4, 5, 6\}$
$B = \{1, 2, 3, 4\}$
$P(A) = \frac{3}{6} = \frac{1}{2}; P(B) = \frac{4}{6} = \frac{2}{3}$
$P(A \cap B) = \frac{1}{6}$
$\Rightarrow P(A \cup B) = \frac{1}{2} + \frac{2}{3} - \frac{1}{6} = \frac{3+4-1}{6}$

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